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Organocatalytic enantio- and diastereoselective synthesis of trifluoro-ethylamine allenoate derivatives containing axial and central chiralities†

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Herein, we report an example of a stereoselective γ-addition reaction of trifluoromethyl ketimines to 1-alkynyl ketones mediated by an isothiourea, BTM, under mild conditions, which afforded tetrasubstituted allenes with central chiralities in high yields (up to 94% yield), good enantioselectivities (up to 91% ee), and excellent diastereoselectivities (all >20 : 1 dr). In addition, the BTM-catalyzed γ-addition reaction was successfully applied to the gram-scale reaction, and an unexpected benzopyrrolothiazine derivative was successfully converted, albeit racemic.

Axial chiral skeletons are widely found in natural products and drug molecules and are often used as chiral ligands or catalysts in asymmetric catalysis.[1] Axial chirality refers to the stereoisomerization phenomenon (IUPAC) formed by the pairwise non-coplanar arrangement of four groups around a chiral axis, and it can be divided into several categories, such as atropisomerism, chiral allenes, and spirocyclic compounds.[2] Among them, chiral allenes are a special class of molecules with unique reactivity patterns and have found a certain presence in natural products and synthetic bioactive molecules such as medicines (Fig. 1).[1,2] For example, enprostil, which is used to treat duodenal ulcers, is more than 600 times more active than its structural analog (**A**),[3] and its analog, fenprostalene (**B**), is a commonly used vasodilator used primarily to improve circulation and relieve pain.[4] Acalycixeniolides E (**C**), a natural product containing an allene structure extracted from Acalycigorgia species of coral, has antitumor activity.[5] Zemlicka develops nucleoside analogs of adenallene (**D**), a prototype for cytotoxic and antiviral drugs currently in the spotlight.[6] Grasshopper ketone 1 (**E**), isolated from the defense secretions of the large flightless locust *Romalea microtera*, and citroside A, a natural product extracted from Cirsium setosum (**F**), both have anti-inflammatory activity.[7,8] The chiral allene units in the above-mentioned reactive molecules are the key structures where biological activity is observed.[9]

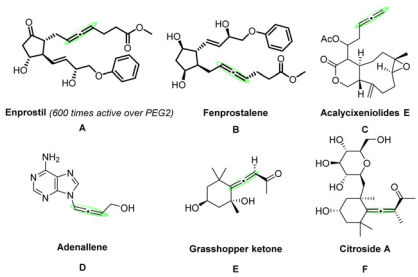


Fig. 1 Allene as scaffolds in natural products and pharmaceuticals.

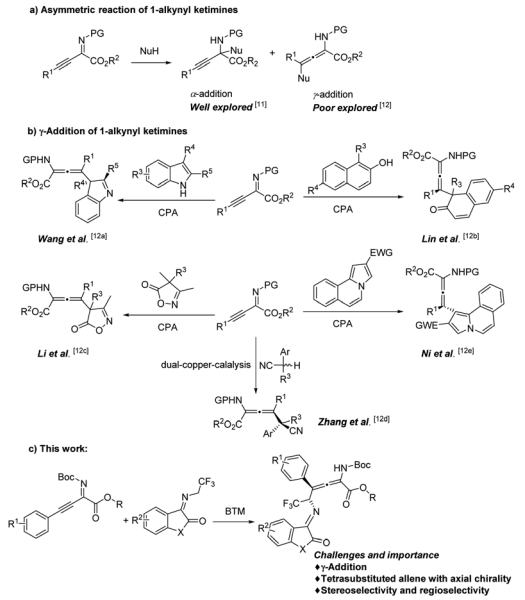
1-Alkynyl ketimines are important synthetics containing multiple active sites (α, β, and γ).[10] In particular, they have been widely used in synthesizing optically active alkyne-based α-amino acid derivatives by catalytic asymmetric α-addition (Scheme 1a).[11] In contrast, selective γ-addition of such substrates has been reported less frequently, but the addition product, alkenylamine, has attracted much attention because of its potent antiretroviral activity and wide range of applications. For example, the first example was reported by Wang's group *et al.*, which developed a chiral phosphate-catalyzed γ-addition of 2,3-disubstituted indoles to 1-alkynyl ketimines (Scheme 1b).[12a] Lin's group investigated naphthol's γ-addition and de-aromatization reactions with 1-alkynyl ketimines.[12b] Li's group developed isoxazol-5(4*H*)-ones with 1-alkynyl ketimines reactions *via* chiral phosphoric acid catalysis.[12c] More recently, Zhang's group asymmetrically synthesized chiral tetrasubstituted

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† Electronic supplementary information (ESI) available. CCDC 2365715 and 2365716. For ESI and crystallographic data in CIF or other electronic format see DOI: https://doi.org/10.1039/d4cc03297g

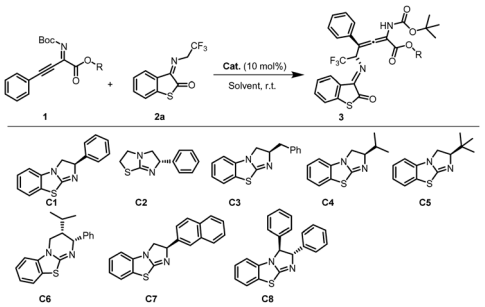
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Scheme 1 Reported α, γ-addition reactions of 1-alkynylketimines (a, b) and this work (c).

Table 1 Optimization of the reaction conditions[a]



Entry	Solvent	Catalyst	Time (min)	Yield[b] (%)	ee[c] (%)
1	Tol	**C1**	50	75	83
2	Tol	**C2**	80	72	80
3	Tol	**C3**	50	65	63
4	Tol	**C4**	30	70	82
5	Tol	**C5**	30	80	90
6	Tol	**C6**	30	50	86
7	Tol	**C7**	40	79	87
8	Tol	**C8**	50	80	83

[a] All reactions were carried out with **1** (0.10 mmol), **2** (0.12 mmol), and catalyst (10 mol%) in Tol (1 ml) at room temperature. [b] The yields of the isolated products are reported. [c] The ee and dr values were determined by HPLC or ^{1}H NMR and ^{19}F NMR.

α-aminoenoates with lateral all-carbon quaternary stereocenters *via* double copper catalysis.[12d] Ni's group recently achieved the pyrrolo[2,1-*a*] isoquinoline C1-site asymmetric allene functionalization reaction by chiral phosphoric acid.[12e] The reported work involves chiral phosphoric acid and Cu-BOX ligand catalytic systems, and no chiral Lewis base has been found to mediate 1-alkynyl ketimine γ-addition to realize the construction of allenes. However, isothiourea BTM catalysts,[13] as an essential class of catalysts in the field of asymmetric organocatalysis in recent years, are widely used due to their powerful catalytic ability, and the related work has proliferated.[14]

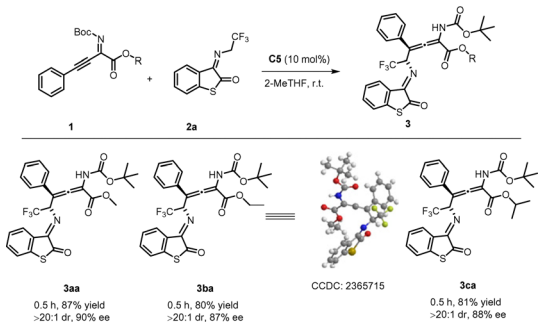
Meanwhile, fluoroalkyl groups are widely used as an important functional group in pharmaceuticals, pesticides, and functional materials.[15] They can regulate the metabolic stability, esterophilicity, and bioavailability of drug molecules. Studies have shown that adding trifluoromethyl to bioactive molecules improves membrane permeability.[16] Therefore, the doping of trifluoromethyl has attracted wide attention, and the technology of trifluoromethyl synthesis has significantly been developed in recent years.[17]

Inspired by these precedents and recent advances in allene synthesis research,[18] as well as the interest in asymmetric organocatalytic construction of fluorine-containing compounds,[19] we herein envision a concept of asymmetric γ-addition of trifluoromethyl ketimine to 1-alkynyl ketimine mediated by isothiourea BTM catalysts to synthesize axial chiral tetrasubstituted allenes bearing trifluoromethyl groups with controlled regioselectivity and stereoselectivity, and the methodology succeeds in remedying the dilemma of having a single catalyst in the area of allene construction and providing a strong rationale for the asymmetric construction of allene compounds containing both central and axial chirality.

To verify the feasibility of our hypothesis, 1-alkynyl ketimine (**1a**) and trifluoromethyl ketimine (**2a**) were chosen as the model substrates. First, various organocatalysts were evaluated, and the reaction progressed well but with poor enantioselectivity (moderate ee or racemic; see ESI† Table S1). Fortunately, the reaction could also proceed smoothly when switching to the isothiourea catalyst **C1** (10 mol%), giving the γ-addition product **3aa** not only in 75% yield, but also with an intrinsically improved enantioselectivity, ee of 83% and dr >20 : 1 (Table 1, entry 1). Importantly, these results confirmed that the BTM catalyst could be effective in this transformation. To improve the yield and enantioselectivity, we first screened the BTM (Table 1). As expected, both yield and enantioselectivity were improved when the catalyst structure was appropriately adjusted (entries 2–8, Table 1), and finally, **C5** was determined to be the most suitable catalyst, which resulted in the γ-adduct **3aa** in 80% yield, 90% ee and >20 : 1 dr (entry 5, Table 1) in 30 minutes. It is noteworthy that further adjustment of the solvent significantly increased the yield (to 87%) while the ee value was unaffected (to 90% ee), and finally, changing the temperature and adding molecular sieves did not improve the reaction results (see ESI,† Table S2). Therefore, the optimal reaction resulted in the γ-addition product **3aa** in 87% yield and 90% ee with 2-MeTHF as a solvent for 25 min at room temperature mediated by catalyst **C5**.

Subsequently, the scope of the substrates was investigated under the optimized conditions established. As listed in Scheme 2, the 1-alkynyl ketimines protected by different alkyl groups reacted smoothly with **2a** to afford the products **3ba** and **3ca** in good yields (80–81%), perfect diastereoselectivities (>20 : 1 dr) and slightly reduced stereoselectivities (87–88% ee) compared to **3aa**. When R was a *tert*-butyl group, the reaction system was a mixture, and the target product (**3da**, Scheme S1, ESI†) was not successfully isolated. Unfortunately,

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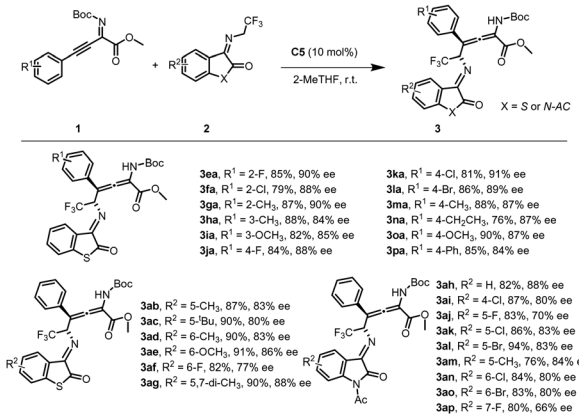
Scheme 2 All reactions were carried out with **1** (0.10 mmol), **2** (0.12 mmol), and catalyst **C5** (10 mol%) in 2-MeTHF (1 ml) at room temperature. The yields of the isolated products are reported. The ee and dr values were determined by HPLC or ^{1}H NMR and ^{19}F NMR.

screening the ester group part observed no better results than the model reaction. By single-crystal *X*-ray diffraction analysis, the absolute configuration of **3ab** was unambiguously established as the (3*S*,5*S*) stereoisomers, and those of the other allenes were assigned by analogy.

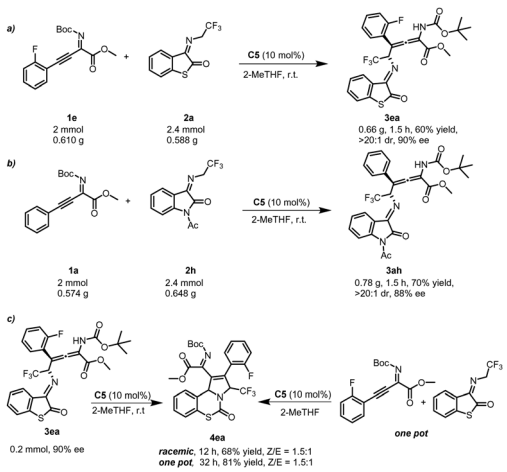
The following examinations were focused on substituents on the benzene ring of 1-alkynyl ketimines, and the results are shown in Scheme 3. Both electron-donating and electron-withdrawing substituents of 1-alkynyl ketimines worked well in terms of stereoselectivity (dr values all $>20:1$). The ee values of the *meta*-substituents are significantly diminished compared to the *ortho*- and *para*-substituents. For the same substituent methyl group, its ee is significantly lower in the *meta*-position than in the *ortho*- and *para*-positions (84 *vs.* 90, 87% ee). However, there was no significant difference in the yield (88 *vs.* 87, 88%). Aryl substrates were tolerated well under standard conditions (85% yield, 84% ee). Furthermore, the generality of dipole **2** was evaluated. In general, the enantiomeric excesses of the evaluated substituted ketimines (Scheme 3, **3ab**–**3ag**) were suboptimal and lower than those of the modeled reactions (77–88% *vs.* 90% ee), whereas the yields, reaction times, and diastereoselectivities did not change significantly. In addition to the novel 1.3-dipole above, we studied trifluoromethyl-ketimine derived from *N*-Ac isatin (**2h**). Recently, it has been widely used in asymmetric organocatalysis. Pleasingly, the reaction proceeded smoothly, and after further optimization of the reaction conditions (see ESI,† Table S2), the γ-site addition product **3ah** was obtained in moderate yield (82%) with excellent diastereoselectivity ($>20:1$ dr) and good excess enantiomeric isomerism (88% ee). Notably, an evaluation of the ketimine substrate range revealed that substituents with different electronic properties (electron-withdrawing or electron-donating groups) and positions gave γ-addition products in moderate to excellent yields (**3ah**–**3ap**, 76–94% yield) and perfect dr (all $>20:1$), however moderate ee values (66–88% ee).

Considering the practicality of this method, we scaled up some reactions under optimal reaction conditions. As shown in Scheme 4, the desired products **3ea** and **3ah** were isolated in sacrificial yields (60 *vs.* 85%, 70 *vs.* 82%), respectively, while maintaining enantiomeric purity (90, 88% ee) and dr values ($>20:1$ dr).

Encouraged by the scale-up reactions above, we investigated the derivatization of **3ea**. Fortunately, it was found that unexpected transformations could be achieved under standard conditions by simply extending the reaction time without changing any conditions or further transformations catalyzed by BTM, as shown in Scheme 4c. The advantage of this synthetic strategy is that the corresponding products can be transformed into valuable benzopyrrolothiazine derivatives **4ea**, whose novel unique frameworks may serve as useful structural moieties in fragment-based drug discovery. Unfortunately, this transformation involves unintended racemization of axial chiral allenes. Given the interesting phenomenon and novelty of the above derivatization, we next evaluated the generalizability of the derivatization, as shown in Scheme S3 (see ESI†), and the single crystal of **4ba** showed that the same reaction pattern had occurred. The results show that this transformation has good generalization, excellent diastereoselectivity, and yields ranging from moderate to high (76–89%). In addition, several other derivative responses were attempted



Scheme 3 All reactions were carried out with **1** (0.10 mmol), **2** (0.12 mmol), and catalyst **C5** (10 mol%) in 2-MeTHF (1 ml) at room temperature. The yields of the isolated products are reported. The ee and dr values were determined by HPLC or ^{1}H NMR and ^{19}F NMR.



Scheme 4 The gram-scale reactions of 3ea (a), 3ah (b) and derivatisation of 3ea (c).

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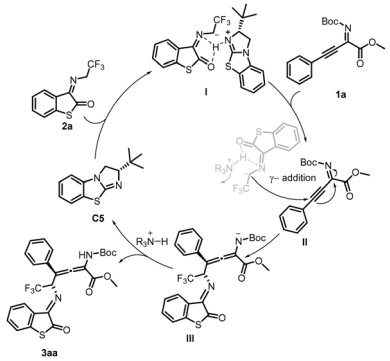


Fig. 2 Proposed mechanism for the reaction.

but unfortunately failed, as detailed in the supporting information document (Scheme S4, see ESI†).

The potential conversion mechanism is shown in Fig. 2: in the presence of tertiary N atoms of the BTM catalyst, the trifluoromethyl ketimine undergoes deprotonation and binds to the catalyst *via* hydrogen bonding, and then the carbon-negative ion attacks the highly electrophilic 1-alkynyl ketimines for γ-site-selective addition, resulting in the generation of the desired axial chiral olefin.

In conclusion, we have demonstrated an asymmetric γ-addition reaction of trifluoromethyl ketimine to 1-alkynyl ketimines with isothiourea BTM catalysis, which was successfully applied to synthesize chiral tetrasubstituted allenes bearing a vicinal central chirality in high yields and with excellent stereoselectivities (up to 94% yield, 91% ee and >20 : 1 dr)[20] under mild reaction conditions. Moreover, the scale-up of the reaction and its derivatization proved the synthetic utility of the scheme, and an unexpected benzopyrrolothiazine derivative could be converted smoothly, albeit racemic.

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Data availability

The data supporting this article have been included as part of the ESI.†

Conflicts of interest

There are no conflicts to declare.

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